NPCYF Documentation

February 26, 2026

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Chapter 1: NPCYF Introduction

1.1 Introduction

1.1.1 About NPCYF

Purpose and Vision

The National Platform for Crop Yield Forecasting (NPCYF) is a comprehensive, web-based digital platform developed to support accurate and efficient forecasting of crop yields across India. Built under the FASAL 2.0 initiative of the Department of Agriculture and Farmers' Welfare (DA&FW), NPCYF represents a significant advancement in agricultural intelligence and decision-making capabilities.

Vision Statement:

To establish a robust, AI-powered national infrastructure that enables stakeholders across the agricultural value chain to access timely, accurate, and actionable crop yield forecasts, thereby contributing to food security, agricultural planning, and policy formulation.

Development Partnership

NPCYF has been developed by IDEAS-Institute of Data Engineering, Analytics and Science Foundation, the Technology Innovation Hub (TIH) at the Indian Statistical Institute (ISI), Kolkata. IDEAS-TIH brings together:

- Deep statistical and mathematical expertise from ISI's legacy
- Advanced capabilities in data science, machine learning, and AI
- Experience in building robust, scalable data platforms
- Understanding of agricultural domain requirements and constraints

1.1.2 Key Features

NPCYF is a comprehensive platform that supports the entire data-to-decision workflow:

1. **Multi-Source Data Integration**
   - Seamless ingestion from diverse data sources (meteorological, hydrological, socio-economic, satellite imagery)
   - Support for structured and unstructured data formats
   - Automated data quality assessment and validation
2. **Advanced Analytics and Modeling**
   - Statistical and econometric models (ARIMAX, panel models, regularized regression)
   - Machine learning algorithms (Random Forest, XGBoost, SVM)
   - Deep learning architectures (CNN, RNN, LSTM)
   - Ensemble methods for optimal forecast accuracy
3. **Feature Engineering Framework**
   - Comprehensive feature registry with version control
   - Automated feature calculation and transformation
   - Feature importance analysis and selection

- Reusable feature sets across multiple models

4. **Model Management**
   - Centralized model registry
   - Hyperparameter tuning and optimization
   - Model versioning and comparison
   - Performance tracking and evaluation metrics
5. **Forecasting Engine**
   - Multiple forecasting techniques (point, probabilistic, ensemble)
   - Exogenous variable estimation for future periods
   - Batch processing for large-scale forecasts
   - Confidence intervals and uncertainty quantification
6. **Platform Administration**
   - Role-based access control
   - User and project management
   - Master data management
   - Audit trails and system logging
7. **AI-Powered CoPilot**
   - Natural language interface for queries
   - Intelligent assistance for data operations
   - Automated model recommendations
   - Insight generation and interpretation

1.2 Home Page

1.2.1 Platform Overview

After successful login, users are presented with the Home Page, which serves as the central information hub for the NPCYF platform.

The home page features a clean, organized layout with:

- **Header Section:** Platform branding with NPCYF logo and organizational information
- **Institutional Partners:** Logos of IDEAS-TIH, ISI, National Mission on ICPS, DST, and Ministry of Agriculture
- **Three Main Information Sections:** What's New, Quick Glance, and About NPCYF

1.2.2 Key Information Sections

The Home Page is organized into several informational sections that provide users with a quick overview of platform capabilities and statistics:

What's New

The What's New section highlights recent enhancements and functional updates introduced on the platform. It communicates improvements aimed at optimizing performance, expanding data coverage, and strengthening analytical workflows.

Recent dashboard enhancements focus on improving system responsiveness and analytical usability. Backend data pipelines and query execution mechanisms have been refined to ensure smoother navigation across dashboards, widgets, and reports, even during periods of high usage. These improvements enable administrators and analysts to monitor datasets, models, and forecasts with minimal delay.

The platform has also introduced new dataset categories across multiple agricultural and environmental domains. These additions broaden the scope of analysis by incorporating diverse inputs such as crop yield data, weather indicators, and related variables, thereby supporting more comprehensive modeling and forecasting.

In addition, the model development pipeline has been updated to improve forecast accuracy and streamline training workflows. These updates help reduce model development time while enhancing the reliability and quality of forecasting outputs generated by the platform.

This section highlights recent platform updates and improvements:

Dashboard Enhancements

- Faster loading and improved analytics experience
- The dashboard has been optimized with refined data pipelines and improved query execution, reducing response times during peak usage
- Users can now navigate between widgets, summary views, and detailed reports with minimal lag, ensuring a smoother monitoring and decision-support experience for administrators and analysts

New Dataset Categories

- Multiple new categories added across key sectors, expanding the platform's data coverage and analytical capabilities

Updated Model Pipeline

- Better accuracy and improved training cycles
- Enhanced forecast quality and reduced model development time

Quick Glance

This section provides at-a-glance statistics about platform resources:

Datasets:

- **Categories:** Organized dataset classifications like APY,Rainfall,Temperature,soil moisture,vegetation index etc
- **Datasets:** Total number of datasets available

Feature Sets:

- **Categories:** Feature classification groups
- **Feature Sets:** Total curated feature collections
- **Active:** Currently enabled feature sets

Model Building Configs:

- **Configs:** Model configuration templates like ARIMA,EXO_ARIMAX,RANDOM FOREST etc
- **Config Items:** Total configuration parameters

Models Trained:

- **Total:** Total number of models trained on the platform
- This metric indicates the cumulative machine learning and statistical models developed for crop yield forecasting

Forecasts:

- **Total:** Total forecasts generated
- **Successful:** Successfully completed forecasts
- This section tracks the number of forecast batches executed and their success rate, providing transparency into forecasting operations

About NPCYF

A comprehensive description of the platform:

"The National Platform for Crop Yield Forecasting (NPCYF) is a web-based system developed under the FASAL 2.0 initiative of the Department of Agriculture and Farmers' Welfare to enhance the accuracy and efficiency of crop yield forecasting across India. By combining satellite observations, field data, weather inputs, and AI-driven models, the platform delivers timely, science-based insights to support planning and decision-making at national, state, and district levels. NPCYF aims to strengthen agricultural preparedness, improve resource management, and contribute to long-term food security by giving stakeholders a reliable, accessible, and data-rich forecasting ecosystem."

Other Links

Quick access to related organizations and resources:

- **IDEAS-TIH:** Technology Innovation Hub at Indian Statistical Institute
- **Indian Statistical Institute, Kolkata:** Partner institution
- **NM-ICPS:** National Mission on Interdisciplinary Cyber-Physical Systems
- **DST, GOI:** Department of Science and Technology, Government of India
- **MoA-FW, GOI:** Ministry of Agriculture and Farmers Welfare, Government of India

Contact Us

Platform support and contact information:

IDEAS-Technology Innovation Hub

- **Address:** 6th Floor, R. A. Fisher Bhavan, ISI, 203 Barrackpore Trunk Road, Kolkata 700108, WB, India
- **Email:** office@ideas-tih.org, tih.isical@gmail.com
- **Phone:** +91 6289-351-800

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1.3. Accessing the Platform

1.3.1 Login Page

The NPCYF login page provides secure access to the platform for authorized users.

Login Page Components:

The login screen displays:

- **Platform Branding:** ”Welcome to NPCYF” - National Platform for Crop Yield Forecasting
- **Project Information:** Developed under the FASAL 2 project of the Ministry of Agriculture and Farmers Welfare, Government of India
- **Username Field:** Enter your registered email address or username
- **Password Field:** Enter your secure password
- **Sign In Button:** Click to authenticate and access the platform
- **Footer Information:**
  - © Institute of Data Engineering, Analytics and Science Foundation
  - Indian Statistical Institute, Kolkata
- **Visual Design:** Clean, professional interface with agricultural imagery (wheat field) representing the platform's agricultural focus

1.3.2 Authentication Process

Step 1: Navigate to Login Page

- Open your web browser
- Enter the NPCYF platform URL provided by your administrator
- The login page will be displayed

Step 2: Enter Credentials

- Enter your registered username/email address in the ”Username” field
- Enter your password in the ”Password” field
- Note: Password entry is case-sensitive

Step 3: Access Platform

- Click the ”Sign In” button
- Upon successful authentication, you will be redirected to the home screen
- Your session will remain active for a configured duration

First-Time Login:

When logging in for the first time, you may be required to:

1. Change the temporary password provided by the administrator
2. Complete your user profile
3. Review and accept platform terms and conditions

1.4. Dashboard Overview

The Dashboard page provides structured access to all platform modules and functionalities through organized cards grouped by operational area.

The dashboard is organized into five main sections:

1.4.1 Data Management

The Data Management section provides tools for managing datasets, ingesting data, and generating reports.

1.4.1.1 Datasets

- **Description:** Create and view datasets
- **Purpose:** Manage dataset definitions, metadata, and configurations

1.4.1.2 Ingestion

- **Description:** Upload and process raw data from various sources into the system, or scrape data from external websites and APIs
- **Purpose:** Configure and execute data ingestion pipelines

1.4.1.3 ETL (Extract, Transform, Load)

- **Description:** Extract and Load data into database
- **Purpose:** Process raw data, apply transformations, and load into the platform's database

1.4.1.4 Reports

- **Description:** Generate and view various reports based on the processed data
- **Purpose:** Create reports, export data, and share insights

1.4.2 Feature Engineering

The Feature Engineering section enables users to define, create, and manage features for model building.This section also supports dataset-level transformations, dataset merging, and temporal interval configuration to prepare analytically robust training datasets.

1.4.2.1 Feature Master

- **Description:** Used to define, manage, and standardize all features (variables) that the platform uses for analysis and model building.
- **Purpose:** Feature Master provides a centralized way to define and manage feature metadata along with their generation logic, including temporal intervals, aggregations, and location levels. It ensures consistency and reusability of features across datasets, training feature sets, and models, while offering simple controls to add, edit, activate, reset, or delete features from a single interface.

1.4.2.2 Training Feature Set

- **Description:** A Training Feature Set is a structured collection of selected features, along with their temporal intervals and configurations, prepared specifically for training a model.
- **Purpose:** It groups relevant features from the Feature Master, defines target variables and time windows, and ensures the right inputs are consistently used for model training, enabling reproducible and model-specific feature preparation.

1.4.2.3 Training Set Data

- **Description:** Training Set Data is the generated and materialized dataset created by applying a selected training feature set to specific crops, seasons, years, locations, and source collections[the below boxes are etl history].
- **Purpose:** It converts configured features into ready-to-use training data by generating and materializing datasets, ensuring consistent, reproducible inputs for model training, validation, and analysis.

1.4.2.4 Training Dataset Collection

- **Description:** A Training Dataset Collection is a logical grouping of multiple training datasets ,of a particular category ,created for a specific modeling or experimentation objective.
- **Purpose:** It organizes and manages related training datasets under a single collection, making it easier to reuse, compare, and maintain datasets across different models, experiments, and analysis workflows.

1.4.2.5 Training Dataset Analysis

- **Description:** Training Dataset Analysis is the stage where the generated training datasets are examined using analytical and visualization tools.
- **Purpose:** It helps analyze what has been trained by evaluating data distributions, trends, and relationships among features and the target variable, enabling validation of data quality and readiness before model training.

1.4.2.6 Training Dataset Operations

- **Description:** Training Dataset Operations is a module that enables transformation and merging of training datasets
- **Purpose:** It allows users to clean, reshape, and combine multiple datasets through transformations and joins, helping prepare consistent, analysis-ready datasets while maintaining traceability of all operations before model training.

1.4.3 Model Building

The Model Building section provides tools for defining model configurations, executing batch model training, monitoring execution status, and managing generated model artifacts.

1.4.3.1 Model Building Config

- **Description:** Model Building Config is a structured setup that defines a unique configuration for model training, combining an algorithm, data split, and hyperparameters into a distinct configuration collection.
- **Purpose:**It standardizes and manages different model setup combinations, ensuring each algorithm–hyperparameter–data split combination is uniquely tracked, reusable, and reproducible for consistent model training and experimentation.

1.4.3.2 Model Building

- **Description:** Model Building is the process of executing configured model training batches using selected datasets and model configurations to generate trained models and related artifacts.
- **Purpose:** It enables systematic execution, monitoring, and management of model training runs, captures run details and statuses, and produces model artifacts and evaluation outputs required for validation, comparison, and deployment.

1.4.4 Forecasting The Forecasting section enables users to configure forecasting parameters, estimate exogenous variables, define forecast batches, and execute forecasts using trained model artifacts.

1.4.4.1 Exogeneous Estimation Config

- **Description:** Exogenous Estimation Config defines configuration collections that specify how exogenous (external) features are estimated or projected for use in forecasting.
- **Purpose:** It organizes feature-wise estimation methods for future periods, links them with training dataset collections, and ensures consistent, reusable, and transparent generation of exogenous inputs required for forecast models.

1.4.4.2 Forecast Config

- **Description:**Forecast Config defines the configuration collections that link trained models (artifacts) with target settings such as forecast year and target variable.
- **Purpose:**It standardizes how forecasts are generated by selecting the appropriate trained model artifacts and target horizon, ensuring consistent, reusable, and well-documented forecast setups.

1.4.4.3 Forecast Building

- **Description:** Forecast Building is the process of creating and executing forecast batches using predefined forecast configurations and exogenous estimation settings.
- **Purpose:**It enables controlled execution of forecasting runs, tracks batch runs and results, and produces final forecast outputs for analysis and decision-making.

1.4.5 Admin Management

The Admin Management section provides administrative tools for platform governance and configuration.

1.4.5.1 User Management

- **Description:** Manage platform users and their roles
- **Purpose:** Create user accounts, assign roles, and control access permissions

1.4.5.2 Projects

- **Description:** View and manage projects on the platform
- **Purpose:** Create projects, associate users and datasets, and organize work

1.4.5.3 Master Data Management

- **Description:** Manage and maintain the core master data for the entire platform
- **Purpose:** Define and update reference data (seasons, crops, states, districts)

1.4.5.4 Copilot Config

- **Description:** Manage the configurations for copilot
- **Purpose:** An AI-powered CoPilot chat interface is available across the platform, allowing users to interact using natural language queries and refresh sessions for continued assistance.

Additional Feature: Chat interface - "Hi! How can I help you today?" with a refresh session option for the AI-powered CoPilot assistance.

1.5. User Roles and Responsibilities

NPCYF implements role-based access control (RBAC) to ensure users have appropriate permissions based on their responsibilities.

1.5.1 Role Definitions

1. Super Admin

- **Primary Responsibilities:** Overall system management, user account creation, role assignment, system configuration, master data management
- **Key Permissions:** Full access to all platform features including Admin Management, User Management, Project Management, and Master Data Management

2. Project Owner

- **Primary Responsibilities:** Project oversight and coordination, user management within projects, resource allocation
- **Key Permissions:** Full access to Project Management functionalities; no access to Admin Management module

3. Data Engineer

- **Primary Responsibilities:** Configure and manage data sources, monitor data ingestion, data quality management, ETL operations
- **Key Permissions:** Access to Data Management components (Ingestion, ETL, Datasets) and others but no access to Admin Management

1.6. Important Contacts

Technical Support: - Email: support@npcyf.gov.in - Phone: +91-XXX-XXX-XXXX - Support

Hours: Monday-Friday, 9:00 AM - 6:00 PM IST

Platform Administration: - Email: admin@npcyf.gov.in - For access requests, role changes,

and administrative queries

Training and Onboarding: - Email: training@npcyf.gov.in - For training schedules,

documentation, and learning resources

Feature Requests and Feedback: - Email: feedback@npcyf.gov.in - We welcome your

suggestions for platform improvements

Emergency Contact: - For critical system issues outside business hours - Phone:

+91-XXX-XXX-XXXX (24x7 emergency line)

1.7 Document Information

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Chapter 2: Ingestion of Data

The Data Ingestion module of the National Platform for Crop Yield Forecasting (NPCYF) forms the entry point for all externally sourced datasets used in forecasting activities. As a national-scale analytical system, NPCYF depends on diverse data streams—agrometeorological, hydrological, agricultural production, remote sensing, and environmental datasets—to support scientifically rigorous and timely crop yield predictions.

Data ingestion ensures that datasets obtained from official and authenticated sources follow a standardized structure, undergo system-level validation, and are transformed into formats suitable for downstream processing. This chapter outlines the sources, procedures, metadata requirements, and system interfaces involved in transferring raw data into the NPCYF ecosystem.

2.1 Purpose and Scope of the Ingestion Module

The Data Ingestion module is designed to:

- Enable seamless upload of raw datasets obtained externally.
- Ensure structural and semantic validation of uploaded files.
- Capture essential metadata (spatial, temporal, administrative).
- Maintain a traceable lineage from source file → ingested dataset → ETL loaded dataset.
- Prepare data for subsequent phases such as feature engineering, modelling, and forecasting.

The ingestion pipeline handles multiple data categories including APY (Area–Production–Yield), rainfall, temperature, reservoir levels, groundwater, and other agro-environmental indicators.

2.2 Supported Data Categories and Source Systems

NPCYF integrates data from government-backed, certified national and state-level systems. Each dataset category has specific designated sources:

2.2.1 Agricultural Production and Yield (APY) Data

APY datasets constitute the core agricultural statistics that document crop cultivation area (hectares), production volume (tonnes), and productivity (yield per hectare) across crops, districts, states, and time periods. These datasets are essential for training predictive models and validating forecast accuracy.

Unified Portal for Agricultural Statistics (UPAG)

- **Access URL:** https://upag.gov.in
- **Authority:** Ministry of Agriculture and Farmers Welfare, Government of India
- **Data Acquisition Procedure:**
  1. Navigate to the Data Access or Statistical Reports section from the main dashboard
  2. Select the relevant category (Crop Statistics, Horticulture Production, Land Use, etc.)

3. Choose the specific year(s) and state(s) or all-India aggregation as required
4. Click "Download Data" to retrieve datasets in Excel (XLSX) or CSV format
5. Verify file completeness and header structure before ingestion

- **Update Frequency:** Annual releases with quarterly advance estimates for major crops
- **Coverage:** Comprehensive national coverage with district-level granularity

2.2.1.1 Sample APY Data structure allowed in the platform

STATEWISE-APY DATA FORMAT SORTED BY CROP & SEASON (ALL INDIA)

DISTRICTWISE APY FORMAT SORTED BY CROP AND SEASON (ALL INDIA)

Below are the other data sources…

IndiaStat Agricultural Database

- **Access URL:** https://www.indiastat.com/data/agriculture/agricultural-production
- **Authority:** IndiaStat Private Limited (aggregates official government statistics)
- **Data Acquisition Procedure:**
  1. Access the Agriculture section from the main navigation menu
  2. Select the desired dataset category (Production, Yield, Land Use, Crop-wise statistics)
  3. Click "View Data" to preview the dataset structure and available variables
  4. Use "Download Excel/CSV" option (institutional login credentials may be required)
  5. Document the data source, access date, and version for metadata records
- **Update Frequency:** Regular updates aligned with government releases
- **Coverage:** Multi-year time series with state and district breakdowns

Directorate of Economics and Statistics (DES) Agriculture Portal

- **Access URL:** https://data.desagri.gov.in/website/crops-apy-report-web
- **Authority:** Directorate of Economics and Statistics, Ministry of Agriculture
- **Data Acquisition Procedure:**
  1. Navigate to the "Crops - APY Report" section
  2. Select specific crop(s), year(s), and state(s) using the interactive filters
  3. Click "Generate Report" to compile the customized dataset
  4. Download the generated report in Excel or CSV format
  5. Verify data completeness, particularly for recent seasons that may have preliminary estimates
- **Update Frequency:** Seasonal updates with advance, first, and final estimates
- **Coverage:** Crop-specific data with administrative hierarchy (national → state → district)

2.2.2 Rainfall Data

Rainfall is a critical agrometeorological variable that directly influences crop sowing decisions, growth patterns, water availability, and ultimately yield outcomes. The platform ingests historical and real-time rainfall data at multiple temporal and spatial resolutions.

Primary Data Sources:

India Meteorological Department (IMD) - Customized Rainfall Information System (CRIS)

- **Access URL:** https://hydro.imd.gov.in/hydrometweb/(S(...))/landing.aspx#
- **Authority:** India Meteorological Department, Ministry of Earth Sciences
- **Description:** CRIS provides comprehensive rainfall data with customizable temporal (daily, weekly, monthly, seasonal, annual) and spatial (gridded, district, subdivision, state) aggregations
- **Data Acquisition Procedure:**
  1. Access the CRIS portal using institutional credentials
  2. Select the geographical region (district, subdivision, state, or custom boundary)
  3. Choose the temporal resolution and date range
  4. Generate the rainfall report using the query interface
  5. Download the dataset in the preferred format
- **Update Frequency:** Daily updates for current season; historical archives available
- **Coverage:** Pan-India coverage with district-level resolution since 1901

Data.gov.in - Open Government Data Platform

- ***Access URL:*** *https://data.gov.in*
- ***Authority:*** *National Informatics Centre, Government of India*
- ***Data Acquisition Procedure:***
  1. *Enter search terms such as "district wise rainfall" or "rainfall India" in the search interface*
  2. *Filter results to identify datasets from IMD or Ministry of Earth Sciences*
  3. *Review the dataset metadata including temporal coverage, spatial resolution, and update frequency*
  4. *Click the "Download" button to retrieve CSV or Excel files*
  5. *Example relevant dataset: "Monthly Rainfall Data – District-wise" with multi-year coverage*
- ***Update Frequency:*** *Variable depending on specific dataset; typically annual updates*
- ***Coverage:*** *District and subdivision level with monthly/seasonal aggregations*

IndiaStat Meteorological Data Repository

- **Access URL:** https://www.indiastat.com/data/meteorological-data/annual-rainfall
- **Authority:** IndiaStat (aggregates official IMD data)
- **Data Acquisition Procedure:**
  1. Log in using institutional credentials
  2. Navigate to the meteorological data section
  3. Select the required dataset (state-wise annual rainfall, season-wise rainfall, etc.)
  4. Choose specific years and parameters using the filter interface
  5. Click "Download Excel/CSV" to obtain the structured rainfall dataset
- **Update Frequency:** Aligned with IMD official releases
- **Coverage:** State and district level with seasonal breakdowns

2.2.2.1 Sample Rainfall Data structure allowed in the platform(all india district wise)

2.2.3 Temperature Data

Temperature variables, including minimum, maximum, and mean temperature measurements, significantly influence crop phenology, growth rates, pest incidence, and yield potential. The platform integrates both gridded and station-based temperature datasets.

Primary Data Sources:

IMD Gridded Temperature Datasets

- **AccessURL:** https://www.imdpune.gov.in/cmpg/Griddata/Max_1_Bin.html#
- **Authority:** Climate Research and Services, India Meteorological Department
- **Description:** Provides high-resolution gridded temperature data (1°×1° or 0.25°×0.25° resolution) covering minimum temperature, maximum temperature, and mean temperature with monthly and annual aggregations
- **Data Acquisition Procedure:**
  1. Navigate to the gridded data download section
  2. Select the temperature variable (minimum, maximum, or mean)
  3. Choose the temporal resolution (daily, monthly, annual) and spatial coverage
  4. Download the binary or NetCDF format files
  5. Extract state-wise or district-wise aggregations using spatial processing tools
- **Update Frequency:** Monthly updates for recent data; complete historical archives
- **Coverage:** Pan-India coverage from 1951 onwards with uniform spatial resolution

IndiaStat Temperature Time Series

- **AccessURL:** https://www.indiastat.com/data/meteorological-data/temperature
- **Authority:** IndiaStat (official IMD data aggregation)
- **Data Acquisition Procedure:**
  1. Access the temperature data section after login
  2. Select state-wise monthly/annual temperature series
  3. Specify minimum, maximum, or mean temperature variables
  4. Choose the required years and geographical units

5. Download in Excel or CSV format for direct ingestion

- **Update Frequency:** Aligned with IMD releases
- **Coverage:** State-level aggregations with monthly granularity

2.2.3.1 Sample Temperature Data structure allowed in the platform(all india district wise of both tmin and tmax)

2.2.4 Reservoir Water Level Data

Reservoir storage levels serve as crucial indicators of irrigation water availability, which directly impacts crop production in command areas and influences regional agricultural planning and risk assessment.

Primary Data Sources:

India Water Resources Information System (India-WRIS) / National Water Informatics Centre (NWIC)

- **Access URL:** https://indiawris.gov.in/wris/#/Reservoirs
- **Authority:** Central Water Commission, Ministry of Jal Shakti
- **Data Acquisition Procedure:**
  1. Access the Reservoirs section from the main dashboard
  2. Click on specific state boundaries or individual reservoir markers on the interactive map
  3. View real-time water level, storage volume, capacity percentages, and historical trend graphs
  4. Click "Download Data" (typically located at the bottom of the data table)

5. Select CSV or Excel format and apply filters for state, date range, or specific reservoirs
6. Save the downloaded file with appropriate naming conventions including date of acquisition

- **Update Frequency:** Real-time updates (daily or weekly) during crop season
- **Coverage:** Major and medium reservoirs across India with storage capacity data

State Water Resource Department Dashboards

Several states maintain dedicated water resource dashboards that provide more granular, district-level reservoir data with higher update frequencies:

State	Portal URL	Dashboard Section
Maharashtra	https://wrd.maharashtra.gov.in	Reservoir Live Storage Dashboard with real-time updates
Karnataka	https://waterresources.karnataka.gov.in	Dam Level Information with historical comparisons
Tamil Nadu	https://www.tn.gov.in/department/33	Reservoir Level Data section with capacity analysis
Telangana/Andhra Pradesh	http://irrigation.telangana.gov.in	Reservoir Status with project-wise breakdowns

Data Acquisition from State Portals:

1. Navigate to the respective state portal
2. Access the reservoir or dam level information section
3. Select specific reservoirs or download aggregate state data
4. Download available formats (typically Excel, PDF, or CSV)
5. Standardize format to align with NPCYF requirements before ingestion

2.2.5 Groundwater Level Data

Groundwater levels indicate the availability of irrigation water from wells and borewells, which supports crop cultivation in areas without surface irrigation infrastructure. Groundwater data is particularly important for rain-fed agricultural regions.

Primary Data Sources:

India Data Portal (Indian School of Business)

- **AccessURL:** https://indiadataportal.com/p/groundwater/r/mojs-wris_cgwb_wells_level_changes-plot-aaa
- **Authority:** Aggregates data from Central Ground Water Board (CGWB) and state agencies
- **Data Acquisition Procedure:**
  1. Navigate to the groundwater monitoring section
  2. Select the geographical region and temporal parameters

3. View visualizations and data tables for well level changes
4. Download the underlying dataset in CSV format
5. Document the measurement period and well locations for metadata

- **Update Frequency:** Seasonal measurements (pre-monsoon and post-monsoon)
- **Coverage:** Well-level monitoring network across India

State Groundwater Boards and Departments

Individual states maintain dedicated groundwater monitoring networks with more frequent measurements and district-level granularity:

State	Portal URL	Section/Report Name
Maharashtra	https://gsda.maharashtra.gov.in	Groundwater Level Status with district reports
Tamil Nadu	https://wrd.tn.gov.in	Groundwater Monitoring Data with well inventories
Karnataka	https://groundwater.karnataka.gov.in	Groundwater Level Reports with seasonal analysis
Telangana	http://groundwater.telangana.gov.in	Water Level Reports with trend analysis
Gujarat	https://gujwater.gujarat.gov.in	Groundwater Monitoring with quality parameters

Data Acquisition from State Portals:

1. Access the specific state groundwater board portal
2. Navigate to the monitoring data or level status section
3. Select district, block, or monitoring well locations
4. Download seasonal reports or raw measurement data
5. Ensure consistency in measurement units (meters below ground level) before ingestion

Before ingestion, users must ensure that all downloaded files meet system requirements regarding header format, data completeness, and structural uniform.

2.3 Data Ingestion tutorials

2.3.1 APY DATA

→CREATING DATASET:

To create a new dataset, open the Datasets tab.Under Create Dataset, we add a new dataset category name(for example,*title as apy2 ,description-upag_testing,data category-APY)*→ click Create Dataset **.**

→DATA INGESTION

To ingest data, go to the Data Ingestion tab → select the dataset that was previously created → enter the required metadata such as the state name, starting year, and header range*(as shown in the image as an example)* → upload the data file from your local system → and wait for the system to confirm a successful upload.

(NOTE: if multi-header say 2 rows are header then, 1-2 should be the input)

Once complete, the dataset is ready for ETL operations.

→DATA ETL

The ETL process in NPCYF is used to load validated data files into the central database.

To perform ETL, open the **Dashboard → ETL** tab → select the required dataset from the dropdown menu (for example, *Dataset - Apy2*) → choose the dataset file (for example, *Crop-Profile_State-Wise*) → and specify the data format (for example, *Advance Estimates*).

Next, select the appropriate APY type and provide an ETL title (for example, *soyabean_kharif_96-24*) → then click **Load into DB** to begin the loading process.

After loading into DB one can view their database.

→DATA REPORTS

Note:data reports are only available for apy datasets

To view summaries of ingested and loaded data, open the Data Reports section → review the dataset status, validation details → and export reports if required.

2.3.2 Rainfall data

→CREATING DATASET:

To create a new dataset, open the Datasets tab.Under Create Dataset, we add a new dataset category name(for example,*title as all india dist wise avg rainfall_2024 ,description-testing,data category-RAINFALL)*→ click Create Dataset

→DATA INGESTION

To ingest data, go to the Data Ingestion tab → select the dataset that was previously created → enter the required metadata such as the state name, starting year, and header range*(as shown in the image as an example)* → upload the data file from your local system → and wait for the system to confirm a successful upload.

(NOTE: if multi-header say 3 rows are header then, 1-3 should be the input)

→DATA ETL

The ETL process in NPCYF is used to load validated data files into the central database.

To perform ETL, open the **Dashboard → ETL** tab → select the required dataset from the dropdown menu (for example, *Dataset - all india dist wise rainfall_2024*) → choose the dataset file (for example,rainfall_district_Daily_2000_2023_updated (1)) Next, select the appropriate APY type and provide an ETL title (for example, *rainfall dataset 2000-2023)*→ then click **Load into DB** to begin the loading process.

After loading into DB one can view their database.

2.3.3 Temperature data

→CREATING DATASET:

To create a new dataset, open the Datasets tab.Under Create Dataset, we add a new dataset category name(for example,*title as TEMPERATURE,description-temperature dataset,data category-TEMPERATURE)*→ click Create Dataset

→DATA INGESTION

To ingest data, go to the Data Ingestion tab → select the dataset that was previously created → enter the required metadata such as the state name, starting year, and header range*(as shown in the image as an example)* → upload the data file from your local system → and wait for the system to confirm a successful upload.

(NOTE: if multi-header say 3 rows are header then, 1-3 should be the input)

→DATA ETL

The ETL process in NPCYF is used to load validated data files into the central database.

To perform ETL, open the **Dashboard → ETL** tab → select the required dataset from the dropdown menu (for example, *Dataset -Temperature*) → choose the dataset file → Next, select the appropriate APY type and provide an ETL title (for example,min temp daily districtwise 1996-2024*)*→ then click **Load into DB** to begin the loading process.

After loading into DB one can view their database.

2.4 Dataset registration and project association

2.4.1 Dataset Registration

This section defines the dataset category under which external data files will be ingested.
Here, we specify the dataset title, description, and data category such as APY, Rainfall, Temperature, Groundwater, etc.

Registration Procedure:

1. Navigate to the **Datasets** tab from the main navigation menu

2. Click on **Create Dataset** to initiate the registration workflow
3. Enter the **Dataset Title** following naming conventions
4. Provide a detailed **Dataset Description** including source attribution
5. Select the appropriate **Data Category** from the dropdown menu
6. Click **Create Dataset** to save the registration

2.4.2 Project Association

Each dataset must be linked to a project before it can be ingested.
Projects are created under Admin Management and allow users to organize datasets by workflow.

Creating project

To create a project, go to Admin Management → Projects. Under Your project→Create Project → enter the project name and a brief description → click Add New Project to save it.

After the project is being created under Associate/Dissociate, a dataset can be linked or removed from any project.This ensures access control and structured dataset management.

2.5 Data ingestion workflow

This tab enables users to upload external data files and define the metadata needed for the platform to interpret them correctly.Here, we select the dataset created earlier and provide required metadata such as state name, starting year, and header range.
If multi-row headers are present, the header range must include all header rows (e.g., 1–2 or 1–3 as shown in examples). Users then upload Excel or CSV files obtained from authenticated sources.Upon upload, the system performs validation of headers, formats, and data consistency.
If the file passes validation, it becomes available for ETL operations.

Technical Insights
NPCYF follows a modern data lakehouse architecture to manage data efficiently.For further details refer to the appendix.

2.6 Data etl (Extract-Transform-Load)

This tab loads the validated data files into the central NPCYF database.

Here, users select the dataset, choose the ingested file, specify the data type (such as APY Advance Estimates), and provide an ETL title.
Once configured, selecting *Load into DB* initiates the ETL process.

After the load completes, a preview table is generated, displaying the processed and structured data.
This ensures that variables such as area, production, rainfall, year, and district/state information have been correctly parsed.

After loading into DB one can view their database.

Technical Insights
During the **ETL process**, NPCYF takes raw data from the Data Lake (MinIO) and transforms it into structured, queryable tables in PostgreSQL.For further details refer to the appendix.

2.7 Data Reports

This section provides summary reports specifically for APY datasets.

It displays the status of ingested and ETL-loaded files, verifies the parsed header structure, and presents completeness and validation summaries.
Users may review the report to identify issues or export it for documentation.

This enables improved data quality monitoring for APY-related forecasting workflows.

To view summaries of ingested and loaded data, open the Data Reports section → review the dataset status, validation details → and export reports if required.

Chapter 3: Feature Engineering

3.1. Introduction to Feature Engineering

Feature engineering refers to the process of transforming raw data into relevant and meaningful input features standardised to be used by the machine learning models to learn useful patterns and then make accurate predictions. This involves various methods, such as: null value treatment, garbage value treatment, sanity checks, encoding, scaling, selection of the most relevant data, discarding insignificant data etc.

The NPCYF Feature Engineering module handles these tasks in six steps, mentioned chronologically as follows:

1. Feature Master
2. Training Feature Set
3. Training Set Data
4. Training Dataset Collection
5. Training Dataset Analysis
6. Training Dataset Operations

3.2. Purpose and Scopes

3.2.1 Purpose of Feature Engineering

- **Model Accuracy and Performance**
- **Data Quality**
- **Interpretability**
- **Overfitting Reduction**
- **Model Training Efficiency**

3.2.2. Scope of Feature Engineering

- **Feature Creation**
- **Feature Transformation**
- **Feature Selection**
- **Missing Value Treatment**
- **Encoding**
- **Scaling and Normalization**

3.3. NPCYF Feature Engineering Steps

3.3.1. Feature Master

This section is, for all intents and purposes, perhaps the most important and valuable section of NPCYF's Feature Engineering module. Because, this section contains the comprehensive masterlist of all possible features that could be deemed meaningful and/or necessary to build any kind of model for the purpose of any kind of prediction and/or forecast using the NPCYF platform.

This is necessary, because it is quite useful to have a well-structured, well thought out, and carefully curated masterlist of features always accessible at hand. On one hand, this rids the end users without admin privilege

of the hassle and headaches of having to start from scratch and create their own feature masterlist - they can simply select any subset of features already available in the Feature Master to create their own, model-specific, feature set for training and prediction.

On the other hand, for the end users *with* admin privilege, it is also beneficial to have access to a pre-existing masterlist of features, so as to gain an overall idea about what kind of features, along with what kind of temporal, spatial intervals and aggregation methods could prove to be useful if added to the Feature Master.

To know how to insert a new feature into the Feature Master, please refer to the *tutorial*.

3.3.2. Training Feature Set

Not all features are equally important for all models and/or all predictions. Nor are they equally useful. Thus, it is necessary to create specific feature sets to train different models for different prediction purposes.

The Training Feature Set section of NPCYF's Feature Engineering module has been designed to help the user not only create model-specific training feature sets but also edit, reset and/or delete them if necessary. This section also makes it easy to keep track of all the different feature sets created to train different models for different prediction purposes.

The user can do so by first defining and describing the feature set they want to create - for the sake of convenience - and then adding as many features from the features available in the Feature Master as wanted/needed to that feature set.

Technical Insight
The most common techniques used to select useful features for a model are - Filter Methods, Wrapper methods, Embedded Methods and Dimensionality Reduction methods. For Further Details, refer to Appendix II

3.3.3. Training Set Data

Simply specifying a training feature set to obtain desired predictions and/or forecasts is not sufficient. In order to train models, actual data must be generated that strictly adheres to the structure and semantics defined by the selected feature set.

The generation of such a dataset involves dynamically constructing the appropriate SQL query based on the chosen feature set and dataset-specific filters (such as crops, seasons, locations, and time range), followed by executing that query. Upon execution, the resulting dataset is materialized as a SQL view for further querying and can optionally be exported or stored in external storage formats such as CSV or Iceberg tables.

The *Training Set Data* section of NPCYF's Feature Engineering module fulfills this role. Each time a training feature set is selected and dataset-specific parameters are provided, this section generates exactly one concrete dataset derived from that specific feature set under the specified filters.

Technical Insight
The "Generate" button, when clicked, takes into account the filters chosen (e.g. crop, year), and then generates the sql query which will be executed to create the dataset abiding by the feature and filter specifications of the selected feature set.
The "materialize" button, when clicked, then executes that generated sql query to create a concrete dataset and then store it in the form of an SQL view and MinIO URL so that the user can later on see the result if desired and/or export it in the form of csv file(s).
For further details, see Appendix II

3.3.4. Training Dataset Collection

While a *Training Set Data* represents a single concrete dataset generated from a specific training feature set and a fixed set of filters, real-world model development and experimentation often require working with multiple related datasets together. These datasets may represent different time ranges, geographic regions, crops, seasons, or experimental splits (such as training, validation, and testing datasets).

The *Training Dataset Collection* section of NPCYF's Feature Engineering module serves as an organizational and management layer over individual training datasets. It allows multiple *Training Set Data* entities to be grouped together under a single logical collection, enabling them to be treated as a coherent unit for downstream model training, evaluation, and experimentation.

Technical Insight
A training dataset collection does not generate new data on its own. Instead, it references and aggregates already-generated training datasets, preserving their individual definitions while providing a higher-level structure for experiment management, dataset versioning, and reproducibility. Each collection can contain one or more training datasets, with enforced uniqueness to prevent accidental duplication within the same collection.

3.3.5. Training Dataset Analysis

A crucial step of data science is to use statistical and visual tools to analyse the available data to understand its main features, discover patterns, notice anomalies, test hypotheses and check assumptions. This is necessary to gain insight about how a data might behave or what kind of modifications might be necessary before it can be fed to a model for training purposes.

The "Training Dataset Analysis" section of NPCYF's Feature Engineering module serves that very purpose. In this section, it's been made possible to filter a dataset by the training feature set it's under and the training dataset collection it's part of, and then fetch its analysis metrics.

It's also possible to visualize the analysis metrics in terms of various graphical representations, and hence get a further idea of the relationships between the features within the datasets.

Technical Insight
The process of analysing a dataset before feeding it to train a model is called Exploratory Data Analysis. It involves summarizing data, visualizing relationships, identifying errors, and generating new questions, ultimately guiding further analysis and decision-making.

3.3.5. Training Dataset Operations

It is sometimes necessary to modify the feature columns of a dataset to make it suitable for model training and/or prediction/forecasting.

Various operations can be performed on different columns based on their nature, requirements, and relationship with each other. The "Training Dataset Operation" section of NPCYF's Feature Engineering module helps perform those operations.

This way, it's made certain on the platform that the datasets are suitable for moving forward with and to train models to get accurate predictions/forecasts.

3.4. Tutorial - Feature Engineering

3.4.1. Tutorial - Part 1: Training Feature Set

Click on "Training Feature Set" on the dashboard, which will bring you to the "Training Feature Set" page.

3.4.1.1. Step 1 - Definition Define your training feature set by naming and describing it. Fill in the "Name" and "Description" form, check the "is Active" checkbox, and then click on the Add button to add your new training set definition.

.

Upon success, a pop-up will appear to let the user know that the new definition for the training feature set has been added to the table.

You have now successfully defined your training feature set.

3.4.1.2. Step 2 - Selecting Features for the Training Feature Set Upon clicking the "Feature" option, we arrive at the corresponding section where we must select features for our training set.

From the "Select Training Set" drop-down list, select your newly defined training set.

From the "Select Category" drop-down list, choose the feature category you want.

Upon selecting your desired feature category, a third drop-down list will appear. The "Select Feature" drop-down displays all features available in the feature master under the specified category. Select the desired feature from the list, check the "is Target" checkbox if this is your target feature (i.e., the feature you want a prediction for).

After selecting the feature, click on the "Add" button, and the feature will be added to your training feature set and will be visible in the table. A pop-up will also appear upon successful addition.

Add as many features as you want for your training feature set.

3.4.1.3. Step 3 - Temporal Interval for Each Chosen Feature Upon clicking on "Temporal Interval", you'll reach the corresponding section, where you need to specify the time interval of each of the features in your training feature set.

From the "Select Training Set" drop-down list, choose your training feature set.

Upon selecting your training set, a second drop-down list will appear. This "Select Feature" list contains all the features you've chosen from the feature master for your training feature set. Select a feature from this list.

After choosing a feature, three more drop-down lists will appear for you to specify the start and end of the interval range, and the base year.

Specify the start and end of your desired interval range, and the base year. Here, the base year signifies that, for a dataset containing data of a feature from one year to another, whether you want to prioritize the beginning or the end of that interval.

After this, click on the "Add" button to submit the temporal interval for your feature. The interval will be added to the table, along with a pop-up showing the success of the procedure.

Do the same for each feature in your training feature set.

Now, you have completed configuring your training feature set and are ready to move on to the next part.

3.4.2. Tutorial - Part 2: Training Set Data

From the dashboard, if you click on "Training Set Data", it takes you to the corresponding page. We will treat this page, for all intents and purposes, as if it's divided into two partitions. This is purely for the convenience and ease of understanding how this page works.

The first Partition contains a drop-down list, "Select Training Set", a table, and three buttons below it - "Delete", "Generate", and "Materialise".

The second partition has multiple drop-down lists and selectors, and two buttons at the very bottom: "Add" and "Reset".

3.4.2.1. Step 1 - Training Set Selection From the "Select Training Set" drop-down list, choose a training feature set that has been defined and added to the platform in the previous part, "Training Feature Set".

Upon selecting a training feature set from the list, all the datasets generated for that training set as of yet will appear in the table.

In the next step, we will show how to generate a new dataset catered to the chosen training feature set and add to the table.

3.4.2.2. Step 2 - Data Selection In the bottom partition, fill up all the forms from top to bottom, and left to right, with all the desired values/ranges/choices suitable for your model and forecasting.

Finally, click the "Add" button to add the specified dataset to the table. Upon successful addition, a pop-up message will appear informing of the completion of the process.

3.4.2.3. Step 3 - Dataset Generation, Materialisation, and Viewing Now that the dataset specification has been added, come back to the table and select the newly added training set data.

Now, this training set data needs to be generated by putting together all the feature columns from various scraped datasets available on the platform. Click on the "Generate" button, and a pop-up will appear informing you that the generation has started in the background.

After the dataset has been generated, click the "Materialise" button. This will make the dataset ready for viewing. A similar pop-up will appear, informing you that the process has started in the background.

After the dataset has been successfully generated and materialised, both the boxes will be checked in the table, and a view button will appear, upon clicking which the dataset will be visible along with an option to download it as well.

Now, you've successfully generated your own training dataset catered to your feature set. We can now move on to the next part of this walkthrough.

3.4.3. Tutorial - Part 3: Training Dataset Collection

On the dashboard, click on "Training Dataset Collection" to reach the corresponding page

.

This page is divided into two sections: "Definition" and "Dataset Items". In the "Definition" section, the dataset collection has to be named and described. And then in the "Dataset Items" section, we gather all the training datasets we want under that collection.

The purpose of having a collection of training datasets is that we can have the same training feature set for different target variables. And for each target variable, we need to generate a new training dataset catering

to the training set feature specifications. Having all of them under one single collection makes accessing them easy and convenient.

3.4.3.1. Step 1 - Definition Fill up the form by adding a suitable name and description for your dataset collection.

After this, click the "Add" button to include your dataset collection in the table. Upon successful addition, a pop-up will appear with a success message.

3.4.3.2. Step 2 - Dataset Items Click on "Dataset Items", and it will take you to the relevant section.

From the "Select Collection" drop-down list, pick the collection you just defined.

From the "Select Training Set" drop-down list, pick your relevant training set. Upon selecting one, a selector list will appear right below, containing all the datasets catering to your specified training feature set. Choose as many datasets as you want to add to your collection.

Click on the "Add" button after picking your datasets, and they will be added to your collection. A pop-up will appear informing you of the successful addition.

You have now successfully created your dataset collection and can proceed with the next part.

3.4.4. Tutorial - Part 4: Training Dataset Analysis

On the dashboard, click on "Training Dataset Analysis" to reach the corresponding page.

Here, we will analyse the datasets under a specific training feature set, fetch the analysis metrics, and then visualize the graphical plottings of the features and their relationships with each other.

3.4.4.1. Step 1 - Selection of the Dataset From the "Select Training Set" drop-down list, choose your desired training feature set. Then, from the "Select Dataset" drop-down list, choose a dataset collection under that training feature set. Upon your selection, all the datasets in that collection will appear in the table below.

Now, click on the "Analyse" button to start the analysis. A pop-up will appear when the analysis metrics have been fetched successfully. Now we are ready to visualize the analysis of our dataset collection.

3.4.4.2. Step 2 - Visualisation Click on "Visualisation", and select your training feature set and corresponding dataset once again. It will appear in the table below.

Now select a row from this dataset and click the "Visualize" button.

Now, a dialogue box will appear with various visualisation options. Select your preferred option and visualize your dataset in a graphical representation.

Now, the analysis of the datasets is complete, and we are ready to proceed with the final part of feature engineering: applying transformation operations on the datasets.

3.4.5. Tutorial - Part 5: Training Dataset Operations

From the dashboard, click on the "Training Dataset Operations" card to reach the corresponding page.

3.4.5.1. Step 1 - Selection of the Dataset Collection From the dropdown list, select a dataset collection.

Upon selection, all the individual datasets available under that collection will be available on the screen.

3.4.5.2. Step 2 - Selection of Dataset From the list of datasets, select a dataset and click on the tab.

3.4.5.3. Step 3 - Selection of Transformation Operation From the "Select a transformation method" drop down list, pick the transformation operation that you want to perform on the columns of your chosen dataset.

Finally, click on the "Preview Transformation" button at the bottom right of the screen to show how your dataset would look like after applying the chosen transformation operation.

If you're satisfied with the results, then click on the "Confirm Transformation" button that appears at the bottom-right of the screen after successful preview generation.

This will actually apply your chosen transformation operation on the selected dataset columns and alter them accordingly.

Chapter 4: Model Building

4.1. Introduction to Model Building

Model building refers to the process of designing, configuring, and training machine learning models so they can learn patterns from data and generate accurate predictions or insights. It involves selecting appropriate algorithms, defining model parameters and hyperparameters, and systematically training models on prepared datasets. Model building also includes experimenting with different configurations, evaluating performance, and refining parameters to achieve optimal results.

The NPCYF Feature Engineering module handles these tasks in six steps, mentioned chronologically as follows:

1. Model Building Config
2. Model Building

4.2. Purpose and Scopes

4.2.1. Purpose of Model Building

- **Pattern Learning:** Enable models to learn meaningful relationships and patterns from historical data.
- **Prediction Generation:** Produce accurate predictions or forecasts for unseen or future data.
- **Decision Support:** Provide data-driven insights to support business and operational decisions.
- **Performance Optimization:** Identify the best model and parameter combinations for optimal results.
- **Automation Enablement:** Allow scalable and repeatable model training and deployment workflows.

4.2.2. Scope of Model Building

- **Algorithm Selection:** Choosing suitable machine learning algorithms based on the problem type.
- **Hyperparameter Configuration:** Defining and tuning parameters that control model behavior and learning.
- **Model Training:** Training models on prepared datasets using different configurations.
- **Evaluation and Comparison:** Assessing and comparing multiple models to identify the best-performing ones.
- **Artifact Generation:** Storing trained models and execution details for reuse, analysis, and deployment.

4.3. NPCYF Model Building Steps

4.3.1. Model Building Config

The Model Building Config section focuses on defining and organizing the parameters required to build machine learning models in a structured manner. This part enables systematic tuning of model configurations before execution. The process begins with the creation of a Config Collection, which acts as a logical grouping of related configurations for a specific algorithm. For example, multiple ARIMA parameter combinations can be grouped under a single ARIMA Config Collection with a clear name and description. This organization ensures clarity, reuse, and easy management of different experimental setups during the model-building process.

Within each Config Collection, Hyperparameters are defined to specify the algorithm name, required parameters, and whether each parameter is mandatory. These hyperparameters form the foundation for creating valid model configurations. The Config Collection Items subsection then defines actual configurations by assigning concrete values to the previously defined hyperparameters for a selected algorithm. Each item represents a unique model setup. Together, these subsections enable controlled experimentation by systematically generating multiple model configurations that can later be used in batch-based model building and execution.

Technical Insight
Hyperparameters are configuration settings defined before training that control a machine learning model's learning process and structure. They influence model complexity, convergence behavior, and overall performance but are not learned directly from data. For further details, refer to Appendix III

4.3.2. Model Building

The Model Building section is responsible for executing the actual creation of machine learning models using the configurations defined earlier. This process is driven through batches that combine dataset collections with config collections. In the Model Build Batch Definition subsection, a batch is created by assigning a name and selecting the appropriate dataset and configuration collections. This setup enables multiple models to be trained simultaneously across different datasets and parameter combinations, ensuring consistency and scalability in model training workflows.

Once defined, the batch is executed through Model Build Batch Execution, where the selected batch is run and its execution status is monitored to verify successful model creation. The Model Build Batch Run Details subsection provides detailed information about each model built, including execution time, date, and associated batch. Finally, Model Build Artifacts stores the generated outputs, such as pickle files for each trained model, allowing models to be retrieved, analyzed, or deployed later with full traceability.

Technical Insight
Evaluation metrics are quantitative measures used to assess how well a machine learning model's predictions match actual outcomes. They help compare models, detect overfitting, and determine how effectively a model generalizes to unseen data. For further details, refer to Appendix III

4.4. Tutorial - Model Building

4.4.1. Tutorial Part 1 - Model Building Config

Go to the Model Building module on the dashboard, and then click on the "Model Building Config" card.

You'll be taken to the corresponding section - Model Building Config.

This section is divided into three further subsections - "Config Collections", "Config Collection Items" and Hyperparameters.

4.4.1.1. Step 1 - Config Collection

In this subsection, the collection is defined by a name and description. It is recommended to name it by the algorithm that'll be used by the models, and describe it as such.

Fill in the "Title" and "Description" in the form below the table.

Then, click the "Add" button to add your newly defined collection to the table of existing collections. Upon successful completion of the process, a pop-up message will appear informing of the addition of the new config collection, and your newly defined collection would be visible in the table as well.

4.4.1.2. Step 2 - Hyperparameters

Click on the "Hyperparameters" button and you'll be taken to the corresponding subsection.

Type in the "Algorithm Name" and suitable algorithm description in the form below the table.

Now click on the "Add Hyperparameters" button on the right side of the form to add your hyperparameters.

Add as many hyperparameters you want, and check the "mandatory" checkbox if any of your included hyperparameters is a must for training your model. After you're done adding hyperparameters, click on the "save" button. It will save your list of hyperparameters under your config collection.

Finally, click on the "Add" button to add your algorithm, along with your specified hyperparameters to the table. A pop-up will appear after the successful addition, as well as your algorithm and hyperparameters appearing on the table.

4.4.1.3. Step 3 - Config Collection Items

Click on the "Config Collection Items" button to go to the relevant section.

From the drop down list, select your config collection that was defined in step 1. Then fill in the title and description in the form. Enter the value for test and train split and select the algorithm defined in step 2. After that, click on the "Hyperparameters" button to assign values to your hyperparameters.

Assign values to your hyperparameters, and then click on the save button.

Finally click on the "Add" button to add your config collection item to the table. A pop-up will appear upon successful completion of the process and your item will be displayed on the table as well.

Now that you've specified your model and hyperparameters, you're ready to move on to the next part and actually build the model.

4.4.2. Tutorial Part 2 - Model Building

Click on the "Model Building" card on the "Model Building" module in the dashboard. It'll take you to the corresponding section.

This section is further divided into four subsections - "Model Build Batch Definition", "Model Build Batch Execution", "Model Build Batch Run Details", and "Model Build Artifacts".

4.4.2.1. Step 1 - Model Build Batch Definition

Click on the "Model Build Batch Definition" button to go to the relevant subsection.

Fill in the "Title" and "Description" section of the form with the name and suitable description of your model building batch. Then from the "Select Dataset Collection" drop down list, choose a dataset collection. Then from the "Select Config Collection" drop down list, select a config collection created in part 1 of this tutorial. Add any comment if necessary.

Click on the add button to include your newly defined batch to the existing list. A pop-up will inform about successful completion of the process and your batch will be visible in the table.

4.4.2.2. Step 2 - Model Build Batch Execution

Click on the "Model Build Batch Execution" button to go to the relevant subsection.

From the "Select Model Build Batch" drop down list, select your batch.

Then click on the “Run” button to run your model on the selected dataset collection.

A popup message will appear upon successful completion of the process and your batch item will also be visible on the table.

4.4.2.3. Step 3 - Model Build Batch Run Details

Click on the “Model Build Batch Run Details” button to go to the corresponding subsection.

Select a model building batch from the drop down list and all the datasets trained under that batch will appear on the table above.

Choose whatever dataset you want to see the details of. To check logs, click on the “View Logs” button and a dialog box will appear displaying the relevant data.

To check what evaluation metrics are being used, click on the “Evaluation Metrics” button, and a dialog box will display all the relevant information.

4.4.2.4. Step 4 - Model Build Artifacts

Click on the “Model Build Artifacts” button to go to the relevant subsection.

From the drop down list, select your model building batch, and all the datasets trained under that batch will appear in the table above.

Pick any dataset of your choice, and then to download artifacts, click on the “Download Artifacts” button. A pop up will appear informing you that it has been downloaded in your local storage.

Chapter 5: Forecasting

5.1. Introduction to Forecasting

Forecasting refers to the process of using historical data, identified patterns, and analytical or predictive models to estimate future outcomes. It enables organizations to anticipate trends, plan resources, and make informed decisions under uncertainty. Effective forecasting combines past performance with assumptions about future conditions and applies suitable models to generate reliable projections. The process often involves configuring forecast horizons, incorporating external influencing factors, and evaluating results for accuracy and consistency. Overall, forecasting serves as a critical bridge between data-driven models and real-world planning and decision-making.

The NPCYF Forecasting module is designed to generate forecasts based on previously built models while incorporating key assumptions and exogenous factors. It allows users to estimate exogenous variables through configurable collections and apply different estimation methods to specific dataset columns. The module then links forecast configurations, including forecast years, with selected trained model artifacts. Using batch-based execution, forecasts can be run across multiple models and configurations simultaneously. Detailed run statuses, execution details, and downloadable forecast results—such as state- and crop-wise projections—ensure transparency, traceability, and effective analysis of forecast outputs.

5.2. Purpose and Scopes

5.2.1. Purpose of Forecasting

- **Future Projection:** Estimate future outcomes based on historical data and trained predictive models.
- **Trend Anticipation:** Identify upcoming trends and patterns to support proactive planning.
- **Decision Enablement:** Support strategic, operational, and policy decisions with forecasted insights.
- **Scenario Analysis:** Evaluate the impact of different assumptions and external factors on future results.
  Planning and Allocation: Aid in effective resource planning, budgeting, and target setting.

5.2.2. Scope of Forecasting

- **Forecast Configuration:** Defining forecast horizons, periods, and related settings.
- **Model Integration:** Connecting trained model artifacts to forecasting workflows.
- **Exogenous Variable Estimation:** Estimating external variables that influence forecast outcomes.
- **Batch Forecast Execution:** Running forecasts across multiple models and configurations efficiently.
- **Result Analysis and Export:** Viewing, analyzing, and downloading forecast results for further use.
  –

5.3. NPCYF Forecasting Steps

5.3.1. Exogenous Estimation Config

In the Exogenous Estimation Config section, the *Config Collection* acts as the foundational layer for estimating exogenous variables required during forecasting. This subsection is focused on defining and organizing the set of configurations that will govern how external (exogenous) variables are projected into the future. These configurations are not the estimates themselves, but structured containers that hold rules, associations, and metadata needed for estimation. By centralizing these definitions, the system ensures consistency, reusability, and traceability across multiple forecasting exercises that depend on the same external variables.

The *Config Collection Items* subsection operationalizes the previously defined configuration collections by assigning concrete elements to them. Here, specific dataset collections and individual datasets—on which predictive models were originally built—are linked to the exogenous configuration. Each variable (column) in the dataset can then be associated with a distinct estimation method. This allows granular control, enabling different estimation techniques to be applied to different variables based on their nature or behavior. The workflow described—adding estimation methods, selecting configuration collections, mapping methods to columns, and saving mappings—ensures a systematic and auditable approach to exogenous variable estimation.

5.3.2. Forecast Config

Within the Forecast Config section, the *Config Collection* defines the core forecasting setup by specifying forecast configurations alongside the relevant forecast years. This subsection establishes the temporal scope and contextual parameters for forecasting, effectively bridging the gap between historical model outputs and future projections. By explicitly defining forecast years within a configuration, the system ensures that forecasts are aligned with the intended time horizon and that all downstream processes operate with a shared understanding of the forecast period.

The *Config Collection Items* subsection connects forecast configurations to the actual predictive models built earlier in the pipeline. This is where the forecast setup is tied to a specific model batch, typically represented through a serialized model artifact such as a pickle file. Selecting the appropriate model ensures that the forecast uses the correct trained logic and parameters. This linkage is critical because it determines which learned relationships are used to generate forecast values, thereby directly impacting forecast accuracy and interpretability.

5.3.3. Forecast Building

In the Forecast Building section, the *Forecast Batch Definition* subsection is responsible for assembling all required components into an executable batch. Here, the Forecast Config is linked with the corresponding Exogenous Config Collection, forming a complete and coherent forecasting unit. This batch-based approach allows multiple forecasts—potentially across different models or configurations—to be defined and managed together. It supports scalability and repeatability, making it easier to run complex forecasting scenarios without manually configuring each run.

The *Forecast Batch Run* subsection handles the execution phase of the defined forecast batches. Once a batch is selected and triggered, the system runs the forecast and tracks its execution status. This includes indicating whether the run was successful or encountered issues. This step is crucial for operational monitoring, as it provides immediate feedback on the health and validity of the forecasting process, enabling users to quickly identify failures and take corrective action.

The *Forecast Batch Run Details* subsection provides deeper visibility into an executed forecast batch. Rather than just indicating success or failure, it exposes detailed information about the run, such as execution metadata and intermediate outcomes. This level of detail supports debugging, auditing, and performance analysis, allowing users to understand how a particular forecast was produced and to diagnose issues if results are unexpected.

Finally, the *Forecast Run Result* subsection presents the actual outputs of the forecasting process. It displays forecasted values segmented by dimensions such as state and crop, making the results directly usable for analysis and decision-making. Additionally, this section allows users to download the forecast outputs and view the estimated exogenous data, ensuring transparency between input assumptions and final predictions. This closes the forecasting loop by making results accessible, verifiable, and reusable.

5.4. Tutorial - Forecasting

5.4.1. Tutorial Part 1 - Exogenous Estimation Config

From the Forecasting module in the dashboard, click on the "Exogenous Estimation Config" card to go to the relevant section.

5.4.1.1. Step 1 - Config Collection

Under the "Config Collection" section, fill in "Title" and "Description" with the title and suitable descriptions of your configuration collection to estimate the exogenous variables.

Then click on the "Add" button to successfully include your newly defined collection to the list. Upon success, a pop-up will appear, and your new collection will be visible in the table as well.

5.4.1.2. Step 2 - Config Collection Items

Click on the "Config Collection Items" button to go to the corresponding subsection.

Fill in the form by selecting appropriate collection sets from the drop down lists and with proper name and suitable description.

Click on the "Assign Estimation Method" button to select appropriate estimation methods for your exogenous variables. A dialog box with a checklist will appear for you to select. Select your desired dataset column(s) as well as required feature filters from the drop down list.

Click on the + button to assign estimation methods. The drop down lists will appear on the task pane for your selection.

Select appropriate choices from the drop down lists and click on the "Save" button.

Check the box and click on the "Save Mapping" button. And the mapping will be saved.

5.4.2. Tutorial Part 2 - Forecast Config

Click on the "Forecast Config" card from the dashboard module, and it will take you to the corresponding page.

5.4.2.1. Step 1 - Config Collection

Fill the form with appropriate title, description and target year.

Click on the "Add" button. Your newly defined collection will be added to the table. A pop-up will appear informing of successful completion of the process.

5.4.2.2. Step 2 - Config Collection Items

Click on the "Config Collection Items" button to reach the corresponding subsection.

Fill up the form with appropriate values, and then click on "Artifacts" button - a checklist will appear for you to select appropriate artifacts.

Select as many artifacts as needed, and then click the "Save" button.

Now, click on the "Add" button to add your collection item to the table. Upon successful completion, a pop-up will appear with a success message.

5.4.3. Tutorial Part 3 - Forecast Building

Click on the "Forecast Building" card from the dashboard to go to the relevant page.

5.4.3.1. Step 1 - Forecast Batch Definition Fill up the form with the appropriate title. Description and comment if necessary. Select a forecast config collection and a exogenous config collection from the respective drop down lists.

Click on the "Add" button. A pop-up will inform you of the successful addition of your definition, and the added definition will be visible in the table above.

5.4.3.2. Step 2 - Forecast Batch Run Click on the "Forecast Batch Run" button to get to the relevant subsection.

Select a forecast batch from the drop down list and all the forecast configs under that batch will be visible in that table.

Select the batch you want to run and then click on the "Run Batch" Button. Your batch execution will start and a pop-up will inform you of the same.

5.4.3.3. Step 3 - Forecast Run Details Click on the "Forecast Run Details" button to go to the corresponding subsection.

From the dropdown menu select a forecast batch to see all the executions under it in the table above.

Select a batch and click on the "View Config" button to see the configuration details.

5.4.3.4. Step 4 - Forecast Run Result Click on the "Forecast Run Result" button to get to the corresponding subsection.

From the dropdown list select a forecast batch to see the results on the table above.

By clicking on the "Download" button, you can download the results if required.

CHAPTER-6 Admin Management

6.0 Introduction to Admin Management

Admin Management serves as the centralized administration layer of the platform, providing authorized users with the tools required to configure, monitor, and govern core system functions. It enables effective management of users and access roles, project oversight, master data maintenance, and configuration of platform-level features such as Copilot settings. This module ensures operational consistency, data integrity, and secure access control, supporting reliable and scalable platform operations.

6.1 Purpose and Scope of the Admin module

The Admin Module is designed to provide centralized control and governance over the platform. It empowers administrators to manage users, data, and configurations efficiently while ensuring security, consistency, and smooth system operations.

Purpose

- To centrally manage platform users, roles, and access permissions.
- To maintain and govern master data used across all modules.
- To oversee and control project-level configurations and workflows.
- To configure and manage system-level features such as Copilot settings.
- To ensure data integrity, compliance, and secure platform usage.

Scope

- User Management: Create, update, and control user accounts and role-based access.
- Project Management: View, manage, and monitor projects across the platform.
- Master Data Management: Define, update, and maintain core reference data.
- Configuration Management: Manage global and feature-specific settings.
- Platform Governance: Enable consistent operations, audit readiness, and administrative oversight.

Overall, the Admin Module acts as the foundation for effective platform governance, ensuring that all users, data, and configurations operate in a controlled, secure, and scalable environment.

6.2 User Management Module

Overview

The **User Management** module under **Admin Management** enables administrators to create, view, and manage platform users along with their assigned roles. This module ensures controlled access, role-based permissions, and secure onboarding of users to the NPCYF platform.

Navigation Path:
Dashboard → Admin Management → User Management

Screen Components

Current Users

- Displays the list of all existing users registered on the platform.
- Allows administrators to review users and verify role assignments.
- Serves as a reference before creating new users to avoid duplication

Create User

- Used to add a new user to the platform.
- Opens the **Enter User Details** form where user credentials and role information are captured.

6.2.1 Current Users

The **Current Users** section displays a consolidated list of all users who are registered on the NPCYF platform. This view enables administrators to monitor and review existing user accounts and their assigned roles.

Table Details

Each row in the table represents a user and includes the following information:

- **Username** – The unique login identifier assigned to the user.
- **First Name** – The user's given name.
- **Last Name** – The user's surname.

- **Email** – The registered email address associated with the user account.
- **Role** – The role assigned to the user, which defines their access level and permissions within the platform.

Purpose

- Provides visibility into all active users on the platform.
- Helps administrators verify role assignments and access levels.
- Supports governance, auditing, and access control management.

This section serves as a quick reference for administrators to ensure that user access is aligned with platform roles and responsibilities.

2.2.2 Create User

The **Create User** section enables administrators to onboard new users onto the NPCYF platform by capturing essential credentials and assigning role-based access. This ensures secure, structured, and controlled access to platform functionalities.

Fields Description

- **Username**
  Enter a unique username that will be used by the user to log in to the platform.
- **Password**
  Set a secure password for the user. The password visibility toggle (eye icon) can be used to view or hide the entered password.
- **First Name**
  Enter the user's first name for identification and official records.
- **Last Name**
  Enter the user's last name.
- **Email**
  Provide a valid email address associated with the user for communication and account-related notifications.
- **Select Role** *(Mandatory)*
  Assign an appropriate role to define the user's access level and permissions within the platform:
  - **Super Admin** – Has complete control over the entire platform, including Admin Management, User Management, Project Management, Master Data Management, and all system-level configurations.
  - **Project Owner** – Has full access to all Project Management functionalities, including creating and managing projects and workflows, but does not have access to the Admin Management module.
  - **Data Engineer** – Has access to all Project Management components required for data ingestion, processing, and analysis, but does not have access to Admin Management features such as user and role management.
- Role selection should be done carefully to ensure proper access control, security, and governance.

Action

- **Add New User**
  Click the **ADD NEW USER** button to create the user. Once created successfully, the user will appear in the **Current Users** list and can log in using the assigned credentials.

This section supports secure user onboarding while maintaining role-based access and platform integrity.

6.3 Projects

The **Projects** module under **Admin Management** allows authorized users to create, view, manage, and organize projects within the NPCYF platform. This section acts as the central workspace for defining projects and associating users and datasets with them.

Navigation Path:
Dashboard → Admin Management → Projects

6.3.1 Your Projects

The **Your Projects** section displays a list of all projects available to the user based on their role and access permissions.

Project List Table

Each project is displayed with the following details:

- **Title** – Name of the project.
- **Description** – Brief description of the project's purpose.
- **Date of Creation** – The date when the project was created.
- **Edit** – Allows modification of project details.
- **Selection Checkbox** – Used to select one or more projects for deletion.

Actions

- **Edit Project**
  Click the edit ( ) icon to update or to modify the project title or description.
- **Delete Selected Projects**
  Select one or more projects using the checkboxes and click **DELETE SELECTED PROJECTS** to remove them from the platform.

6.3.1.1 Create a New Project

The **Create a New Project** section allows users to add new projects to the platform.

Fields Description

- **Title** *(Mandatory)*
  Enter a unique and meaningful name for the project.
- **Description**
  Provide a brief description explaining the objective or scope of the project.

Actions

- **Create Project**
  Click **CREATE PROJECT** to save and create the project. Once created, the project will appear in the **Your Projects** list.

- **Reset**
  Click **RESET** to clear all entered fields.

6.3.2 User – Project Association

The User – Project Association section is used to manage the assignment of users to specific projects within the NPCYF platform. This functionality ensures that users have access only to the projects relevant to their roles and responsibilities.

Purpose

This section enables administrators to associate or dissociate users with projects, thereby enforcing project-level access control and supporting secure collaboration.

Functionality

- **Select Project *(Mandatory)***
  Choose a project from the dropdown list. Once a project is selected, the system allows administrators to manage user associations for that specific project.
- **Associate Users**
  Assign selected users to the chosen project, granting them access to work on that project based on their assigned roles.
- **Dissociate Users**
  Remove users from a project when access is no longer required, ensuring controlled and up-to-date permissions.

6.3.3 Dataset – Project Association

The Dataset – Project Association section allows administrators and authorized users to link or unlink datasets with specific projects. This ensures that each project has access only to the datasets relevant to its objectives, supporting secure and organized data usage.

Purpose

This section is designed to manage dataset accessibility at the project level, enabling controlled data sharing and maintaining data governance across the platform.

Functionality

- Select Project *(Mandatory)*
  Choose a project from the dropdown list. Once selected, datasets can be associated or dissociated for that project.
- Associate Datasets
  Link one or more datasets to the selected project, making them available for use within that project's workflows and analyses.
- Dissociate Datasets
  Remove datasets from a project when they are no longer required, ensuring only relevant data remains accessible.

Purpose and Outcome

The **Projects** module ensures structured project management by:

- Organizing work into clearly defined projects.
- Controlling user and dataset access at the project level.
- Supporting collaboration, governance, and efficient workflow management across the platform.

This module forms the backbone of project-driven operations within the NPCYF platform.

6.4 Master Data Management

The **Master Data Management** module under **Admin Management** is used to define, maintain, and manage core reference data required across the NPCYF platform. This data acts as standardized inputs for multiple workflows, ensuring consistency, accuracy, and uniformity throughout the system.

Navigation Path:
Dashboard → Admin Management → Master Data Management

Overview

Master Data includes essential reference entities such as **Season, Crop, State, and District**, which are commonly used across ingestion, processing, analysis, and reporting modules. Administrators can add, update, delete, or bulk-upload these values from this section.

Components of the Screen

1. Type of Master Data (Dropdown)

Use the **Type of master data** dropdown to select the category you want to manage.

Available options include:

- **Season**
- **Crop**
- **State**
- **District**

The table on the right updates dynamically based on the selected master data type.

Master Data Types

6.4.1 Season

- Displays predefined agricultural seasons such as *Kharif, Rabi, Summer,* etc.
- Used across datasets and forecasting workflows.
- Administrators can add new seasons or remove existing ones if required.

6.4.2 Crop

- Displays the list of supported crops (e.g., *bajra, barley, cotton, cereals,* etc.).
- These values are used during data ingestion and analysis.
- Administrators can add new crops or remove existing ones if required.

6.4.3 State

- Displays the list of Indian states and union territories.
- These values are used as geographical identifiers across the platform.
- Administrators can add new state or remove existing ones if required.

District

Requires **State selection** before displaying district data.

- Once a state is selected, the corresponding districts can be viewed and managed.
- Ensures correct state–district hierarchy and prevents data inconsistency and duplication.
- Administrators can add new districts or remove existing ones if required.

Purpose and Benefits

- Ensures standardized reference data across all platform modules.
- Reduces data duplication and inconsistency.
- Enables controlled updates through authorized administrators.
- Supports scalability through bulk upload functionality.
- Maintains clean hierarchical relationships (e.g., State → District).

6.5 Copilot Configuration

The Copilot Config module under Admin Management is used to define, manage, and monitor AI Copilot configurations within the NPCYF platform. This module allows administrators to control how the Copilot behaves by configuring schemas, prompts, and activation status, ensuring consistent and governed AI-assisted interactions.

Navigation Path: Dashboard → Admin Management → Copilot Config

6.5.1 About the CoPilot

The NPCYF CoPilot is an AI-powered conversational assistant integrated into the platform to provide users with intelligent, context-aware support. It helps users interact with the system more efficiently by answering queries, providing guidance, and assisting with data-related tasks.

Key Features:

- Conversational Interface – Users can interact with the CoPilot through a natural language chat interface, making it easy to ask questions and receive instant responses.
- Context-Aware Responses – The CoPilot uses configured schemas and prompts to understand the data structure and provide relevant, accurate answers based on the platform's datasets and workflows.
- Customizable Behavior – Administrators can define how the CoPilot responds by creating and managing different configurations with specific prompts and schemas tailored to various use cases.
- Session Management – Users can refresh the session at any time to start a new conversation, ensuring clean and focused interactions.

User Interface:

The CoPilot interface features:

- A chat window with a friendly greeting: ”Hi! How can I help you today?”
- A message input field for users to type their queries
- A ”Send” button to submit questions
- A ”Refresh the session” option to restart the conversation

How It Works:

1. Users access the CoPilot from within the platform
2. They type questions or requests in natural language

3. The CoPilot processes the query using the active configuration's schema and prompt
4. It returns intelligent, contextual responses based on the platform's data and defined behavior

The CoPilot Configuration section allows administrators to create, modify, and manage these AI configurations to ensure the CoPilot delivers accurate and helpful assistance aligned with platform objectives and governance standards.

6.5.2 Definition

The Definition section is used to create and manage Copilot configurations.

Configuration List Table

Displays existing Copilot configurations with the following columns

Title – Name of the Copilot configuration (e.g., ”demo1”, ”t1-2”, ”wer”)

- Description – Brief explanation of the configuration's purpose
- Active – Checkbox indicating whether the configuration is currently enabled (checked = active)
- Schema – Defines the data structure or context used by the Copilot (e.g., database schema, table structures)
- Prompt – Instruction that guides Copilot behavior (e.g., ”generate answers for me”, ”You are an intelligent assistant...”)
- Edit – Pencil icon ( ) allowing modification of an existing configuration
- Selection Checkbox – Located on the left of each row for selecting configurations for bulk operations

The table also includes:

- Pagination controls at the bottom (”1 to 7 of 7”, ”Page 1 of 1”)
- Sorting capabilities on each column header
- Visual indication of active configurations with checked boxes in the Active column

This table helps administrators quickly review and manage available Copilot setups.

Create / Edit Copilot Configuration

Below the table, administrators can add or modify Copilot configurations using a form with the following fields **:**

Form Fields:

- Title (Mandatory - indicated by red asterisk *)
  - Enter a unique and descriptive name for the Copilot configuration

  - Example: "demo1", "Data Collection Assistant"
- Description
  - Provide a short description explaining the purpose or use case of the configuration
  - Example: "Assists users with data collection queries"
- Schema (Mandatory - indicated by red asterisk *)
  - Define or review the schema that structures Copilot inputs and outputs
  - The schema field displays database schema information in a structured format
- This defines the database tables, columns, and relationships the Copilot can reference
- Prompt (Mandatory - indicated by red asterisk *)
  - Enter the instruction that determines how the Copilot should respond and behave
  - Example: "You are an intelligent assistant helping users with agricultural data queries. Use the provided schema to answer questions accurately."

Action Buttons:

- FETCH SCHEMA (top right button)
  - Click to automatically retrieve the relevant database schema
  - Populates the Schema field with current platform data structures
  - Ensures the Copilot has access to the latest schema information
- Active (checkbox on the right)
  - Enable this checkbox to make the configuration active
  - Only active configurations are used by the system
  - When checked, the configuration becomes available for use in the CoPilot interface
- ADD (gray button, bottom left)
  - Saves a new Copilot configuration
  - Adds the configuration to the table above
- RESET (white button, center)
  - Clears all entered fields in the form
  - Useful for starting fresh without saving
- DELETE (red button, bottom right)
  - Removes the selected configuration
  - Only available when editing an existing configuration

This configuration allows the Copilot to answer questions about agricultural production yield data by referencing the specified database schema.

6.5.3 Logs

The Logs section provides visibility into Copilot usage and activity.

- Helps administrators monitor execution, usage patterns, and potential issues
- Supports auditing, debugging, and performance tracking

- Enables review of user interactions and Copilot responses

Purpose and Benefits

- Ensures controlled and consistent AI behavior across the platform
- Enables structured prompt and schema management
- Allows easy activation or deactivation of Copilot features
- Supports transparency and governance through logs
- Provides flexibility to create multiple configurations for different use cases
- Ensures the Copilot has accurate, up-to-date schema information through the Fetch Schema feature

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